WHISPER ENERGY

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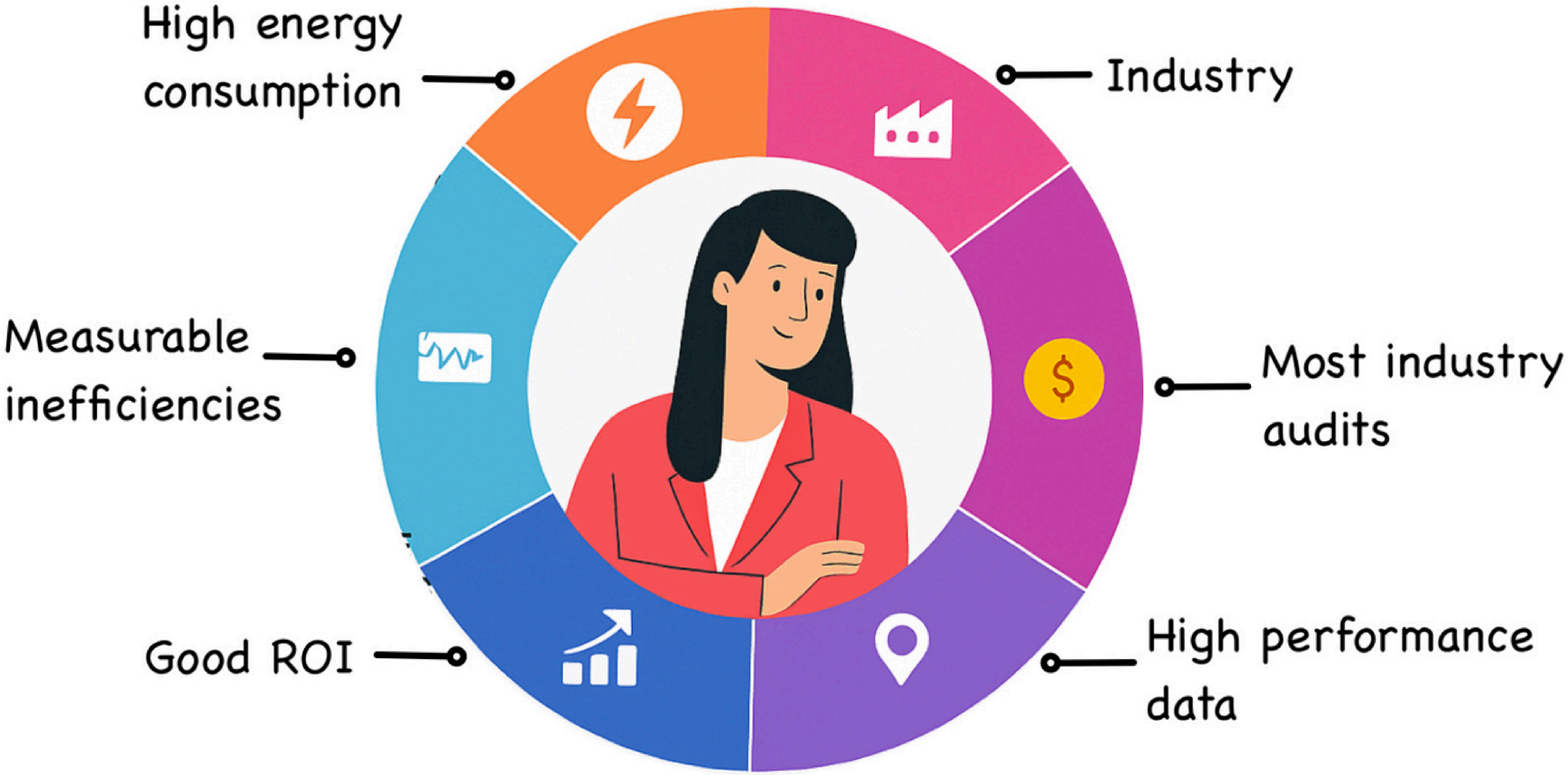
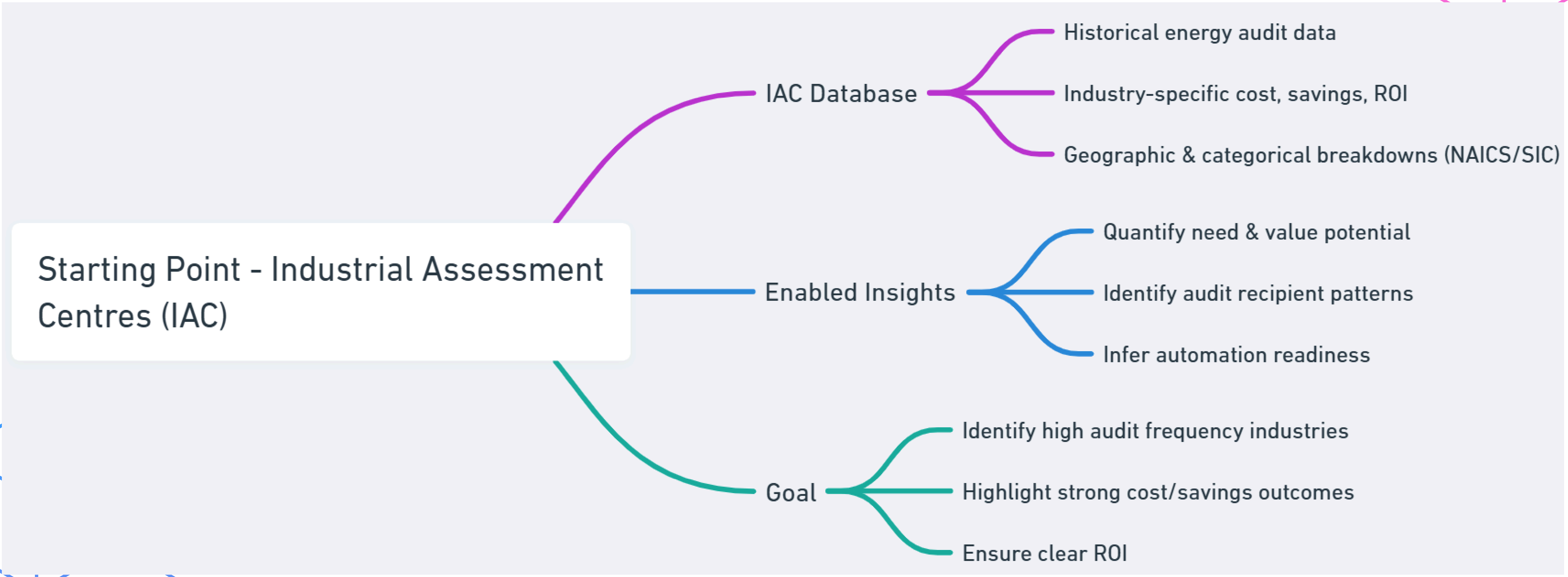
What is an ICP?

- An **Ideal Customer Profile** is a data-backed representation of the best-fit clients for **WHISPER's** sensor automation solutions.

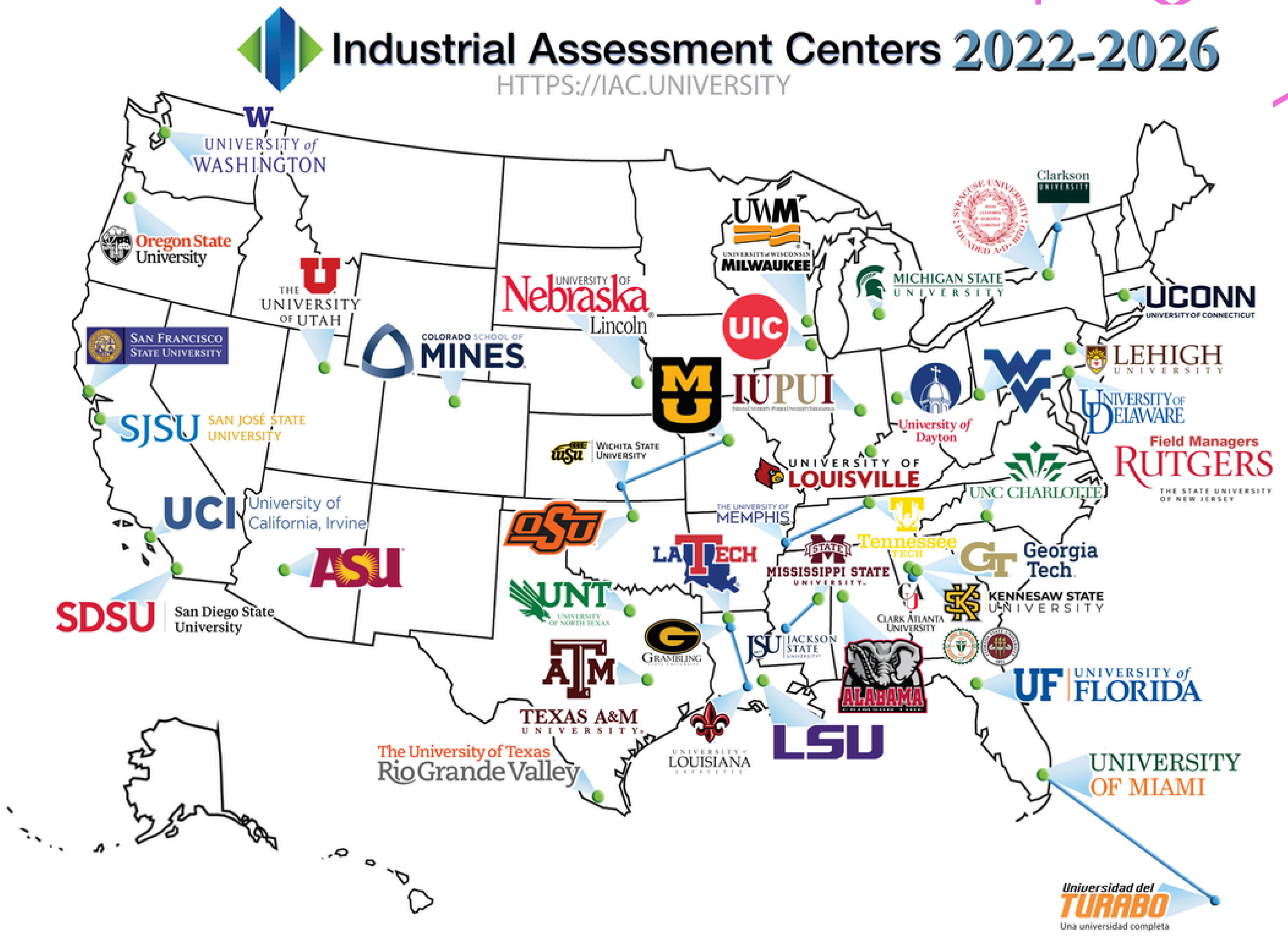


Where did we start?



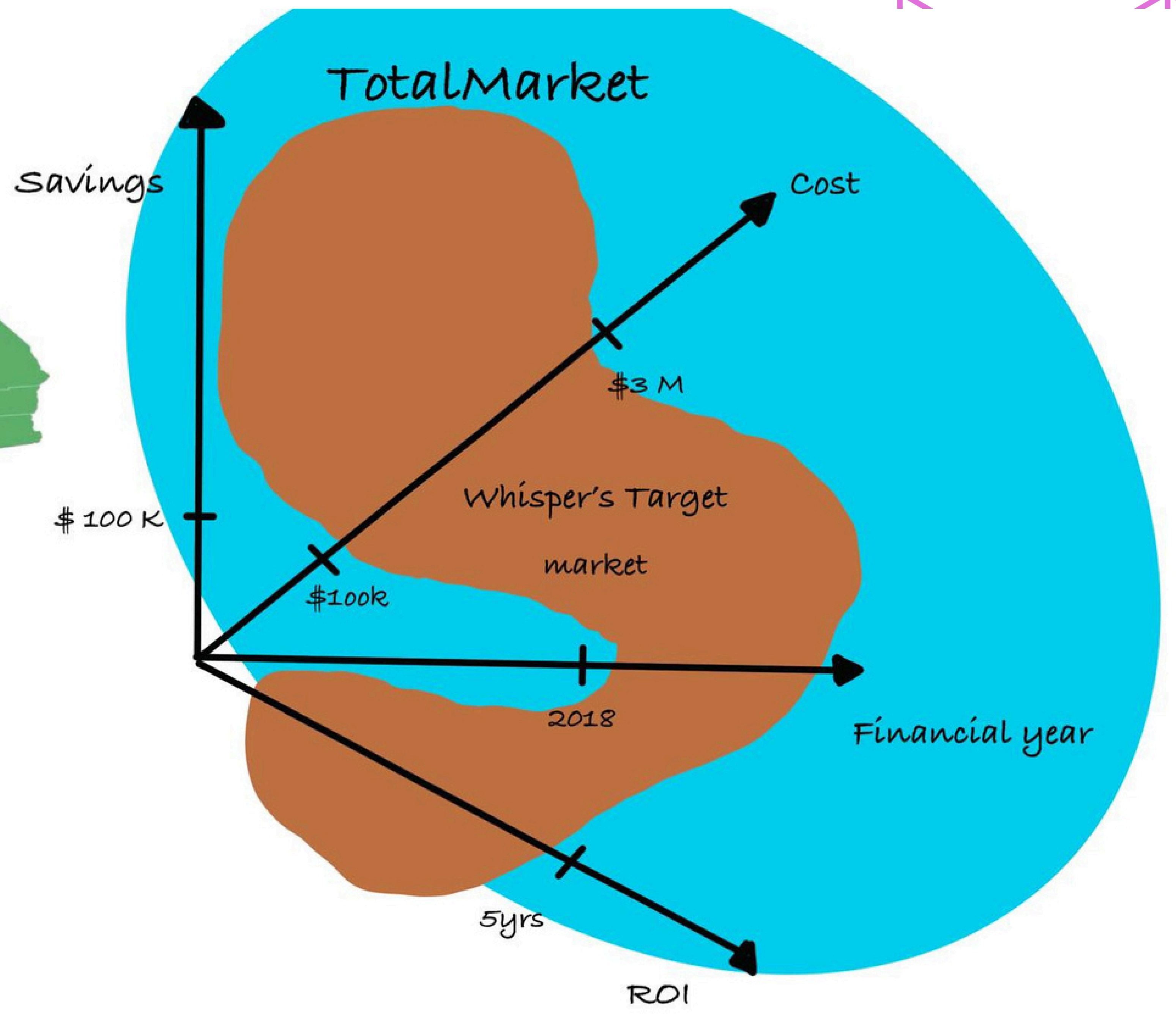
More about IAC

- 15,000+ audit records across all 50 U.S. states
- Covers manufacturing, utilities, facilities, and specialized industries
- Data includes:
  - Number of audits per sector
  - Total audit cost per sector
  - Cost savings recorded
  - ROI metrics (often expressed as payback periods)



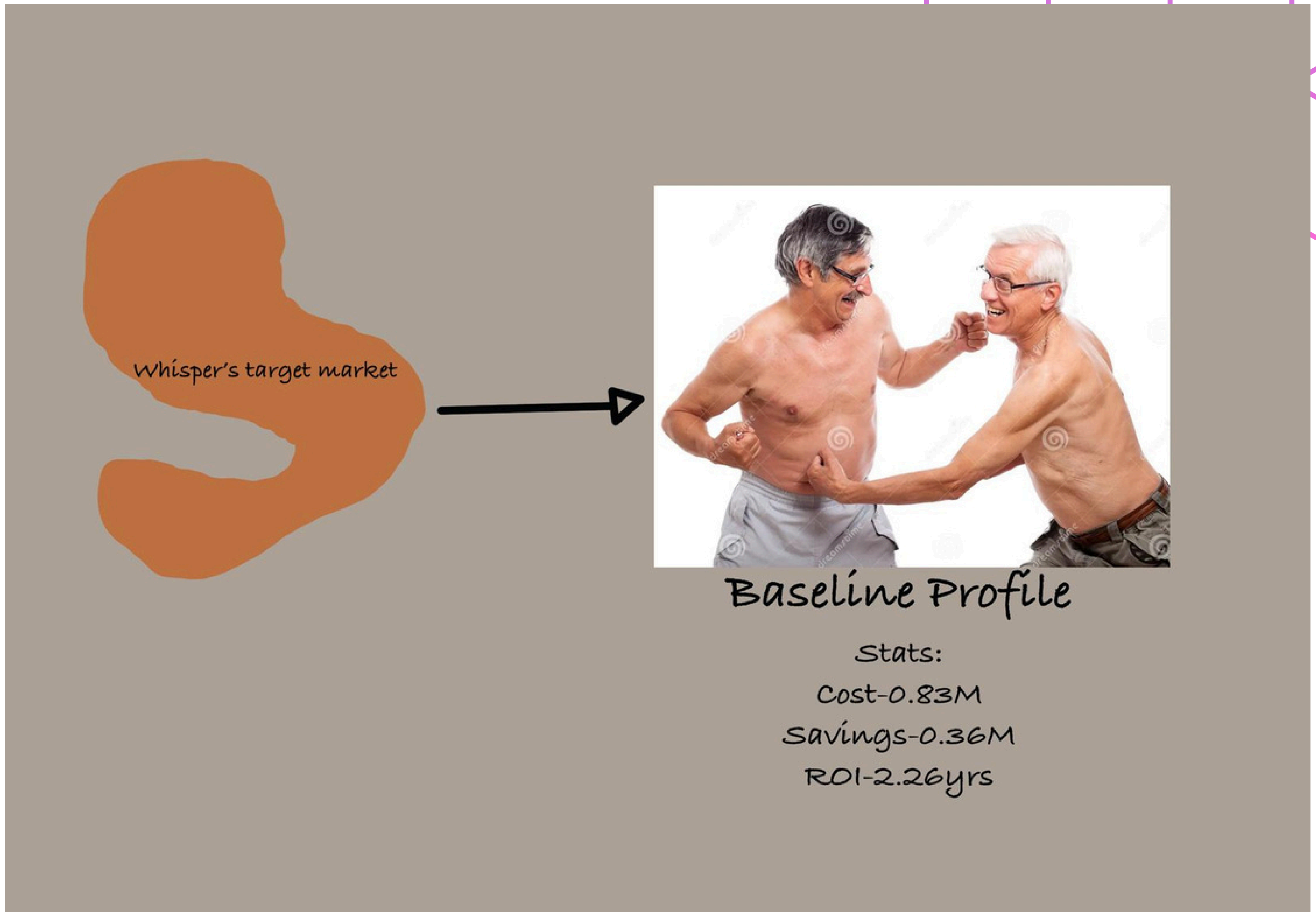
Data Filtering

- When?FY>=2018
- Where?California(pilot)+National(roll-out)
- How much?
  - Electricity $100K - $3M
  - Savings >= $100K
  - ROI <= 5yrs



Baseline Profile

We build our baseline profile with the filtered data and this is the benchmark we compare every industry against.

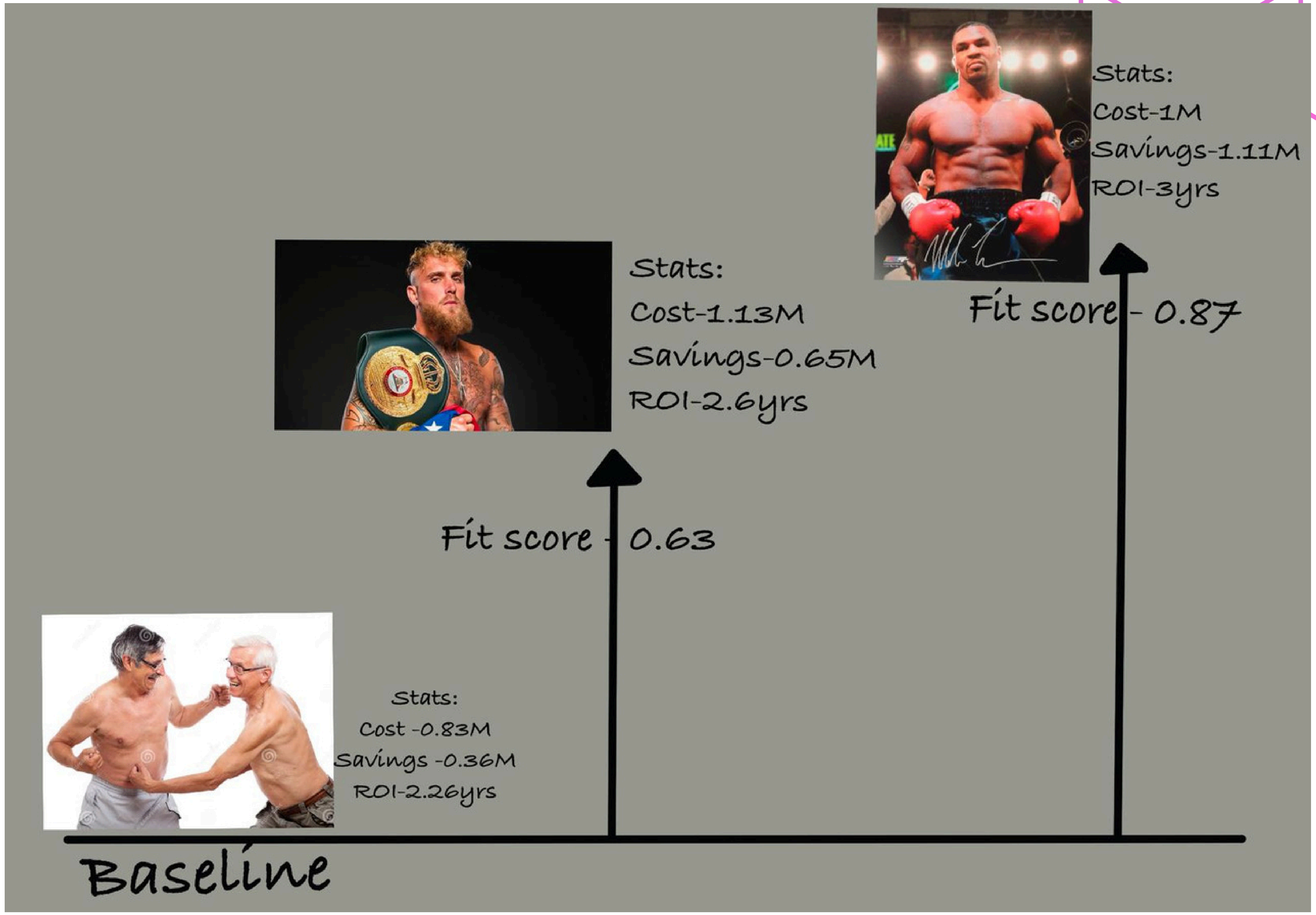


California Baseline

FY	SIC	NAICS	Annual Sales (M)	Employees	Plant Area (M sqft)	Electricity Cost (M)	Impl. Cost (M)	Savings (M)	ROI (yrs)	PRODUNITS	PRODLEVEL	PRODHOURS	NUMARS	num_reccs	Major Group
2022.681818	3396.94697	335277.113636	56.59956	123.189394	0.261109	0.831207	0.737957	0.369121	2.261986	2.885246	3.303976e+07	5451.181818	8.386364	8.386364	33.363636

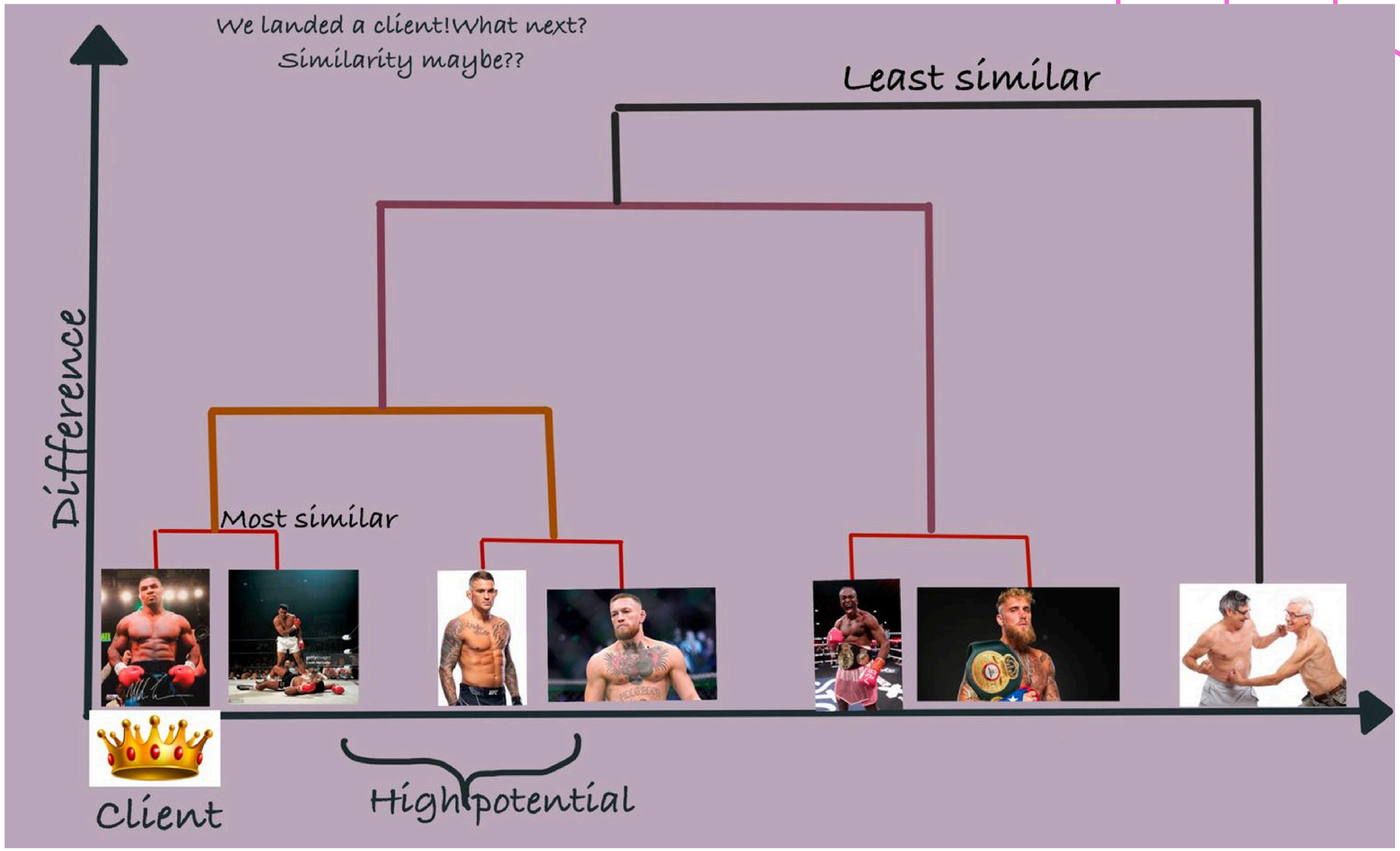
Targeting Strategy

- We compute a Fit Score(-1 to +1) for each industry by considering its deviation from the baseline on audit frequency,annual electricity costs and other key metrics.
- Positive scores reward sectors with high audit counts,big energy bills and huge potential savings.
- Industries with the highest Fit Scores are our top ICP targets.



Multi-Sector and Similarity Analysis

- Once we pick a target SIC and it proves to be fruitful we go on to discover it’s closest peers through methods like Ward’s hierarchical clustering to their standardized metric vectors,then use Pearson correlation to quantify and rank inter-sector similarity.



Web App Integration

- We built a Streamlit dashboard to deliver on-demand sector analytics: per-SIC descriptive statistics, metric distributions with μ/median overlays, and side-by-side comparisons versus the baseline.
- The app also supports multi-industry clustering and Pearson correlation heatmaps, making similarity insights instantly accessible.

Single SIC Comparison

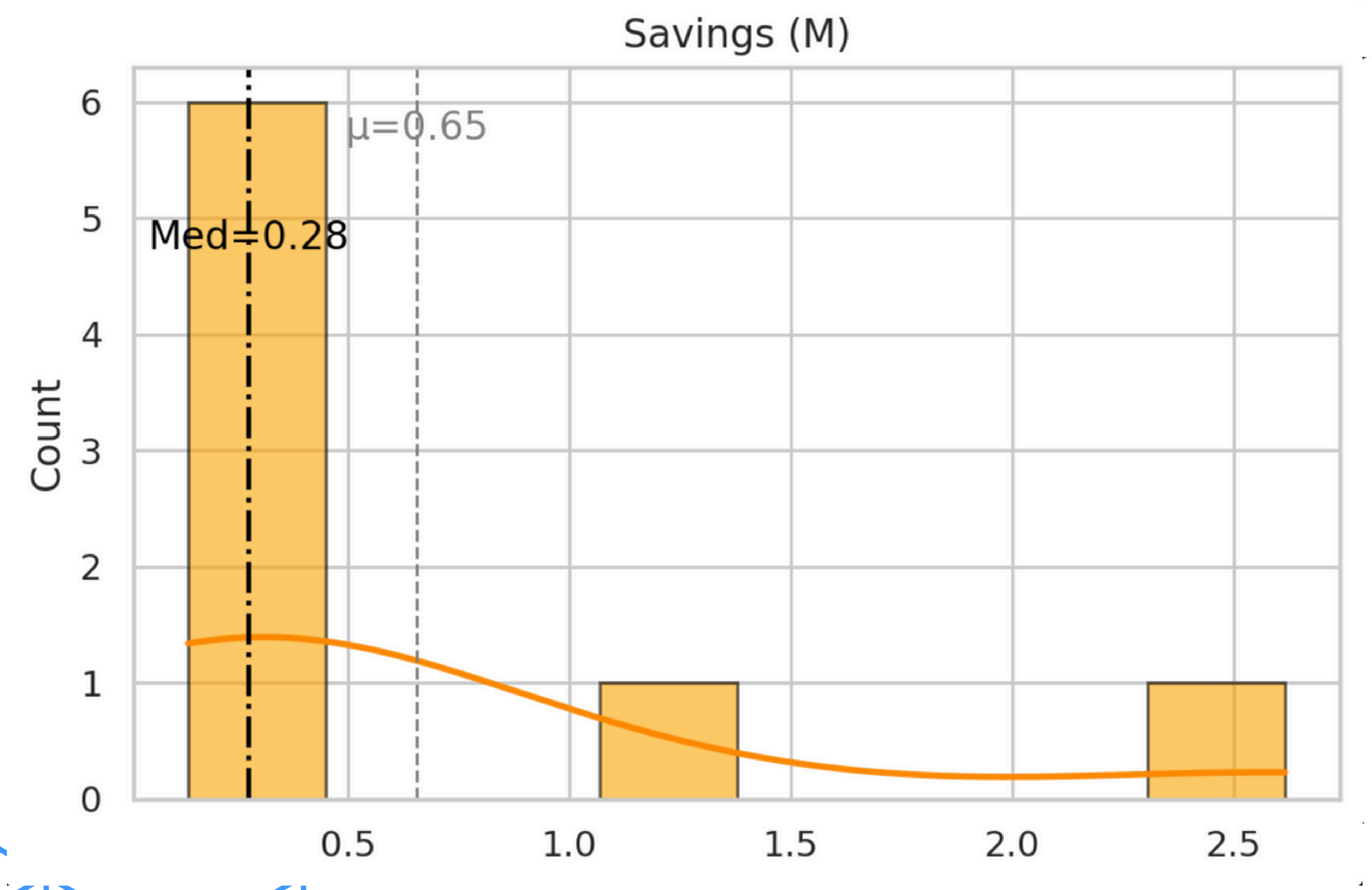
SIC 4952 – Sewerage Systems (8 records)

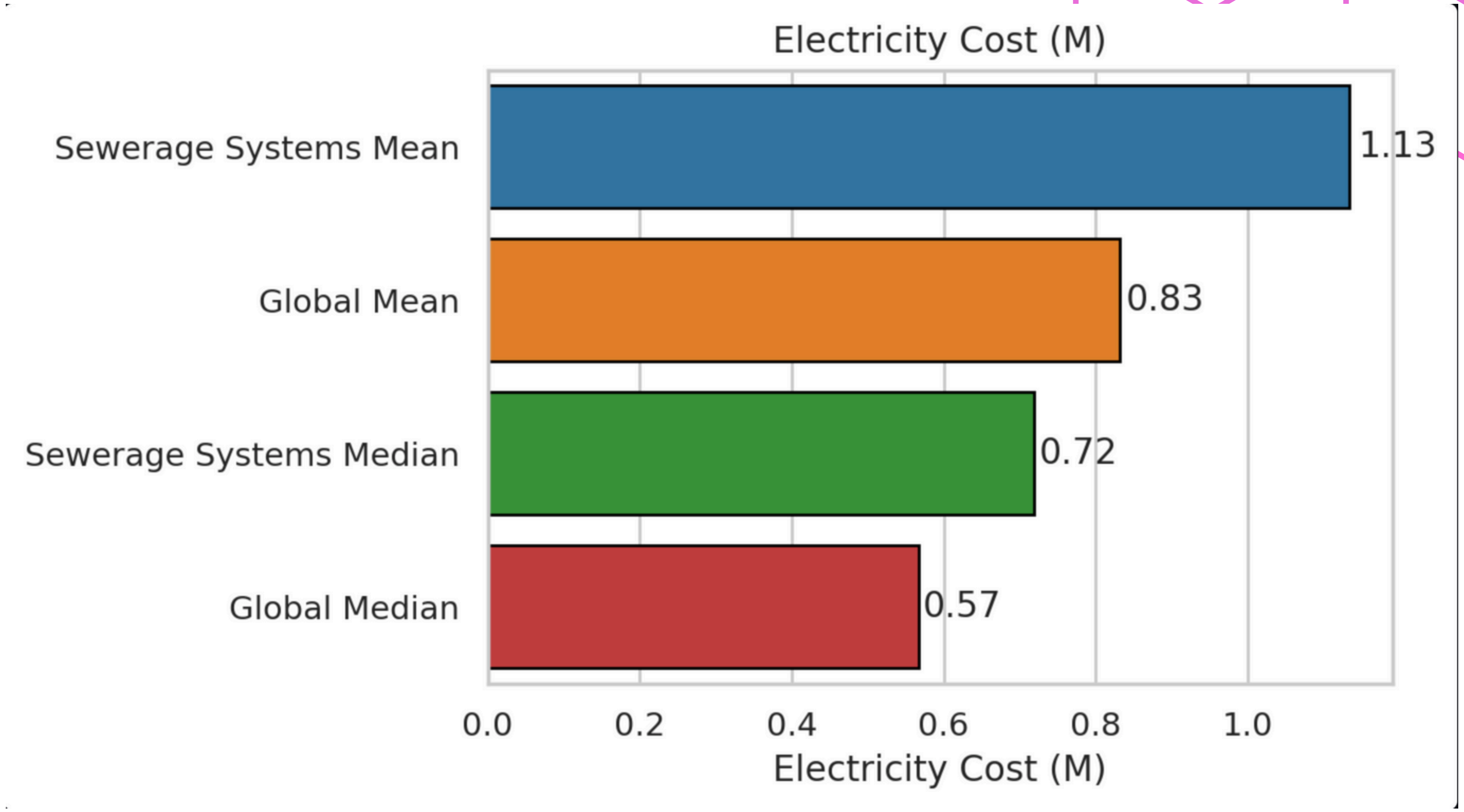
Sector Summary

	count	mean	std	25%	50%	75%
Annual Sales (M)	8.00	11.76	20.60	1.99	3.25	9.06
Employees	8.00	64.38	89.82	16.75	37.50	53.25
Plant Area (M sqft)	8.00	1.92	3.50	0.32	0.71	1.12
Electricity Cost (M)	8.00	1.13	0.84	0.63	0.72	1.20
Implementation Cost (M)	8.00	2.10	3.84	0.52	0.76	1.22
Savings (M)	8.00	0.65	0.85	0.23	0.28	0.58
ROI (yrs)	8.00	2.60	1.15	1.70	2.55	3.33

Sector Descriptive Statistics:

Web App Integration

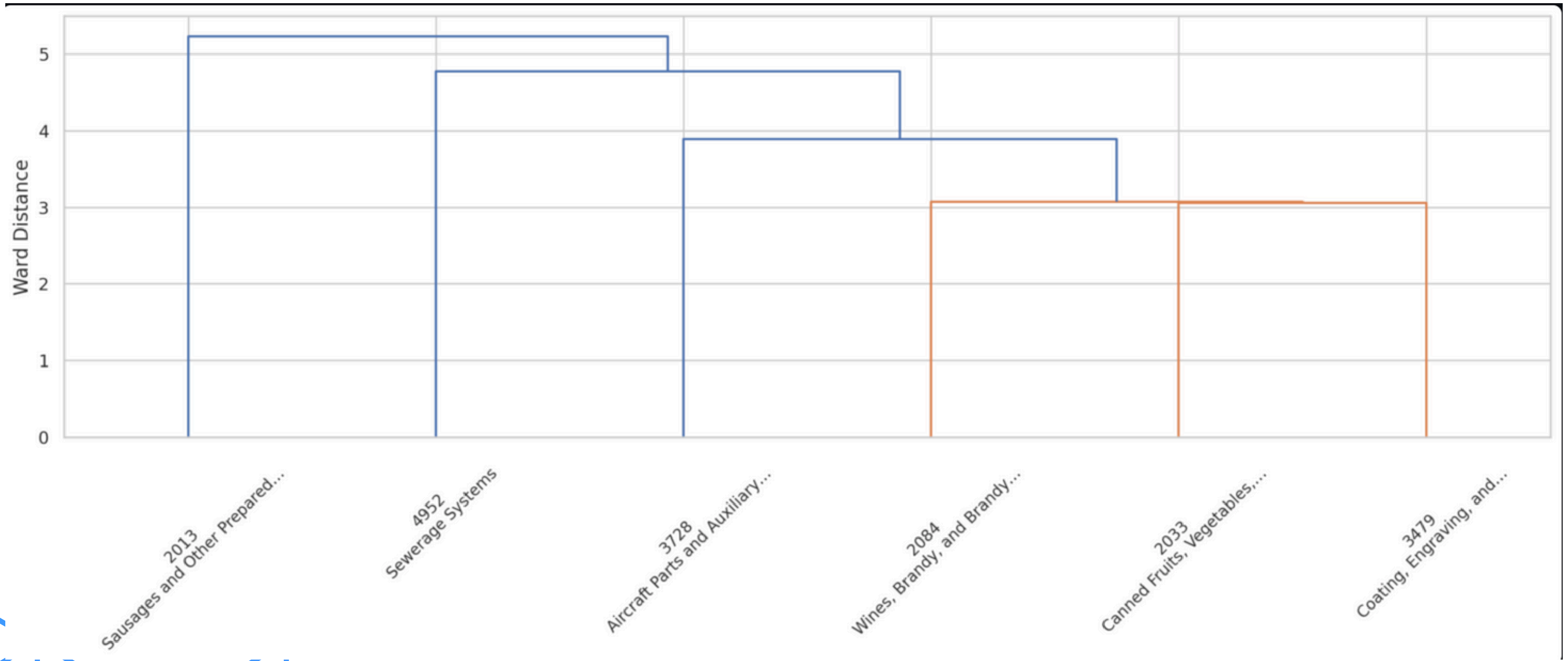




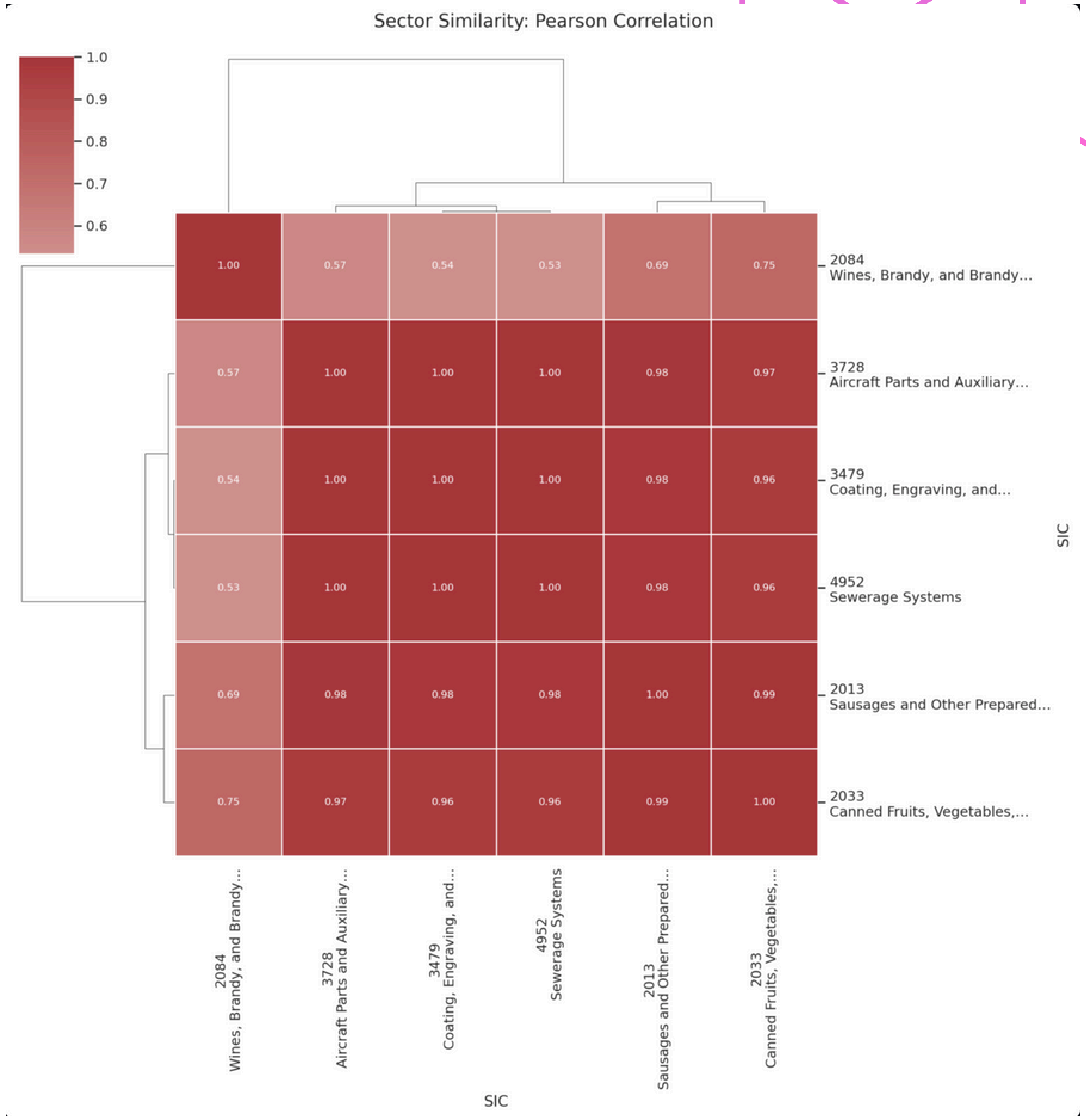
Distribution of metrics:

Sector vs Baseline Comparison

Web App Integration



Ward Hierarchical Clustering:



Corelation Heatmap:

Findings - ICP

State Level (CA)

Industry	Sales	Savings
Aircraft	60 million	1 million

National Level

Industry	Sales	Savings
Sewerage Industry	12 million	0.5 million

State vs. National Analysis: Key Differences

Criteria	California (State-Level)	National-Level
Top Industry	Wines, Aircraft, Sewerage, Coating	Sewerage, Motor Vehicles, Aircraft
Savings Leaders	Wines ($3.4M), Aircraft ($3.2M), Coating ($2.9M)	Sewerage ($78M), Transport Equip. ($25M)
Audit Frequency	High in Sewerage & Wines	Broad across many sectors
Group Strength	Group 20, 34, 49 (Strong overlap in CA)	Group 20 (Consistent), 37 (Emerging), 34, 49

Transformative

Get In Touch

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